

Comparing Statistical Learning Models for Reinforcement Learning

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Abstract

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Declaration

I declare that this thesis was composed by myself and that the work contained therein is my own, except where explicitly stated otherwise in the text.

(Ioan Baker)

...dedication text...

Contents

Abstract	ii
Contents	v
1 Introduction	1
1.0.1 Guideline	1
1.0.2 Structure of Introduction	1
1.0.3 Multi-Armed Bandit Problem	1
1.0.4 Statistical Learning Methods	2
Bibliography	3

Chapter 1

Introduction

1.0.1 Guideline

Reinforcement learning (RL) is a popular type of machine learning, where automated agents learn to act in a way that maximises their rewards in an unknown environment. Standard approaches to RL involve an agent (acting in the environment), a state space (describing all possible states the agent can be in), and an action space (describing the possible actions the agent can take in each state). Human agents are not perfect and consequently may not always pick the theoretically optimal decision, and a good statistical learning algorithm should be able to learn this uncertainty in the decision making process.

This project investigates the applicability of different statistical learning algorithms to A) estimate either the agent's policy or the underlying reward function, and B) estimate the (un)certainty of this policy or the reward function. Depending on the students' interests, this can be achieved either using theoretical considerations, by a numerical study (using e.g. examples from the mujoco library in Gymnasium, see <https://gymnasium.farama.org/>), or by analysis of a real-data set with a clearly defined target objective.

1.0.2 Structure of Introduction

- What is reinforcement learning. How does it compare and differ to supervised/unsupervised learning.
- What is reinforcement learning used for and real examples

In an RL algorithm, an agent (define) seeks to maximise some reward function by taking actions ..

1.0.3 Multi-Armed Bandit Problem

One of the simplest forms of RL is the multi-armed bandit problem in which there is only a single state. The agent's actions $A_t \in \mathcal{A}$ correspond to a realised sample of a reward function R_t but crucially they do not change the environment state. Written formally, given a set of actions $\mathcal{A}$ and number of rounds T , for each $t \in T$, the agent selects an action (arm) $A_t \in \mathcal{A}$ and observes some reward

R_t . This means, if the agent can learn which action's corresponding reward function is the most favorable, it The algorithm used to find this most favourable arm uses the notion of an **action value function** $q_*(a) = \mathbb{E}[R_t|A_t = a]$. The agent will aim to learn the true action value function by interacting with the environment while simultaneously trying to accumulate the most reward possible. This requires a careful balance between **exploitation** where the agent selects the action which is currently assumed to be the most rewarding and **exploration** where the agent selects actions about which it has less information. It is important to consider the explore-exploit dilemma since too much exploitation may lead to more profitable actions not being taken due to lack of knowledge while too much exploration will not allow the agent to reap the rewards from the most profitable actions.

To estimate the action value function, one can simply use the sample average $Q_t(a) = \frac{1}{N_t(a)} \sum_{\tau=1}^{t-1} R_\tau \cdot \mathbb{1}_{A_\tau=a}$ (This is if the reward function outputs binary rewards.) Where $N_t(a)$ is the number of times that action a has been taken at time t , and R_τ is the reward observed at time τ . It should be obvious (or not and prove) that the sample average converges to the true action value as the number of times that an action is taken approaches infinity. $\lim_{N_t(a) \rightarrow \infty} Q_t(a) = q_*(a)$.

1.0.4 Statistical Learning Methods

This dissertation aims to demonstrate how different statistical learning methods can be used for RL, so it is important to

References